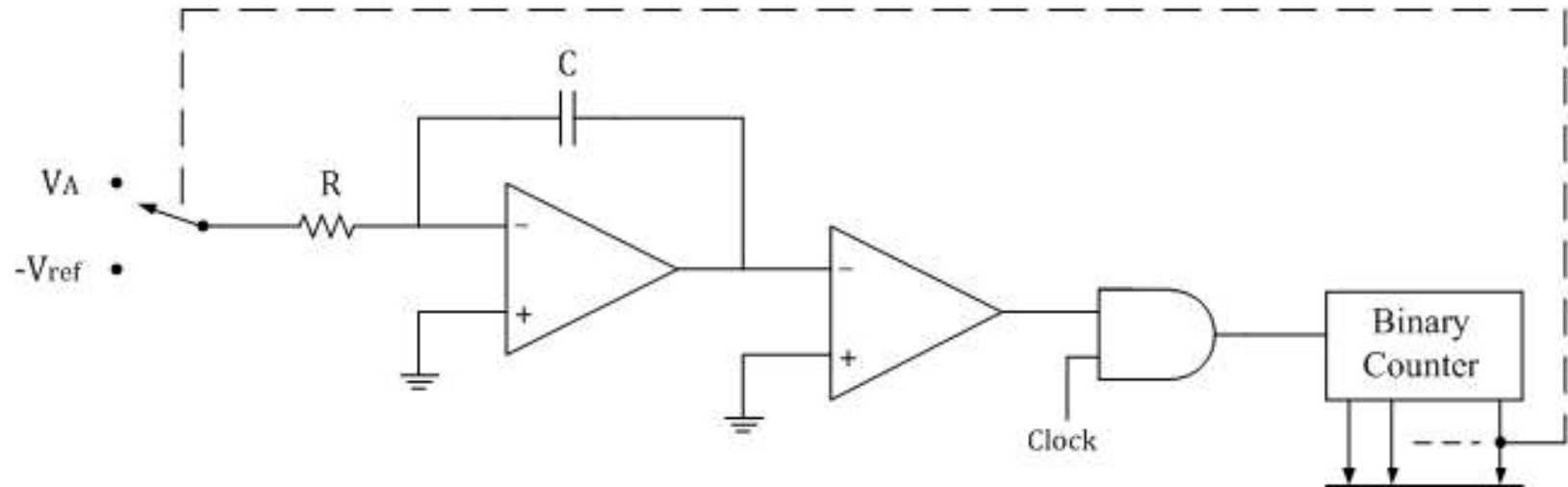


Dual slope ADC



A dual slope ADC uses an integrator for conversion. This provides high resolution with good accuracy but it has a very slow conversion time. So, they are not used to gather data but preferred in places where the signal changes slowly.

- Case 1: When connected to V_A / V_{in} , from time $t < 0$ to $t = T_1$

$$V_o = -\frac{1}{RC} \int_0^{T_1} V_{in} dt$$

$$V_o = -\frac{V_{in} * T_1}{RC}$$

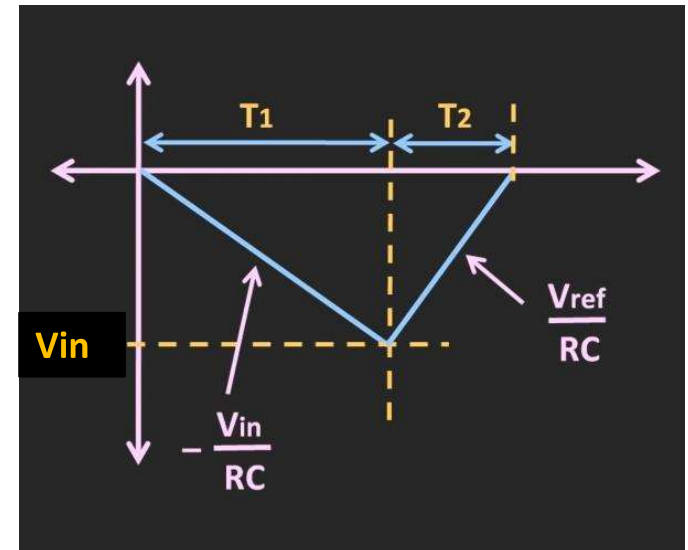
From time $t = 0$ to $t = T_1$, the counter counts from 0 to 2^n and after reaching 2^n , a control signal is sent to the switch and the switch connects to $-V_{ref}$ so the capacitor starts discharging.

$$V_o = -\frac{1}{RC} \int_{t_1}^{t_2} -V_{ref} dt + V_{initial}$$

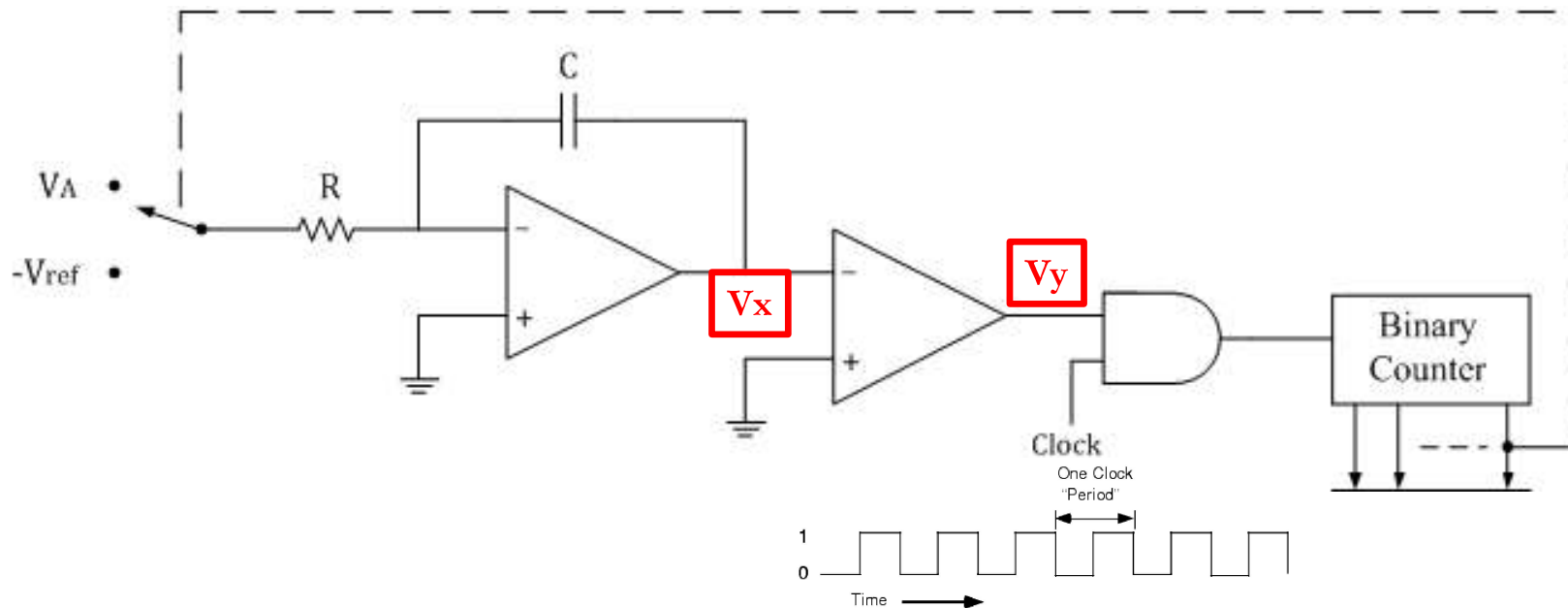
$$V_o = \frac{V_{ref} * T_2}{RC} - \frac{V_{in} * T_1}{RC} = 0$$

$$\text{So, } \frac{V_{ref} * T_2}{RC} = \frac{V_{in} * T_1}{RC}$$

$$T_2 = \frac{V_{in} * T_1}{V_{ref}}$$



Case 1: When connected to V_A / V_{in} , from time $t < 0$ to $t = T_1$

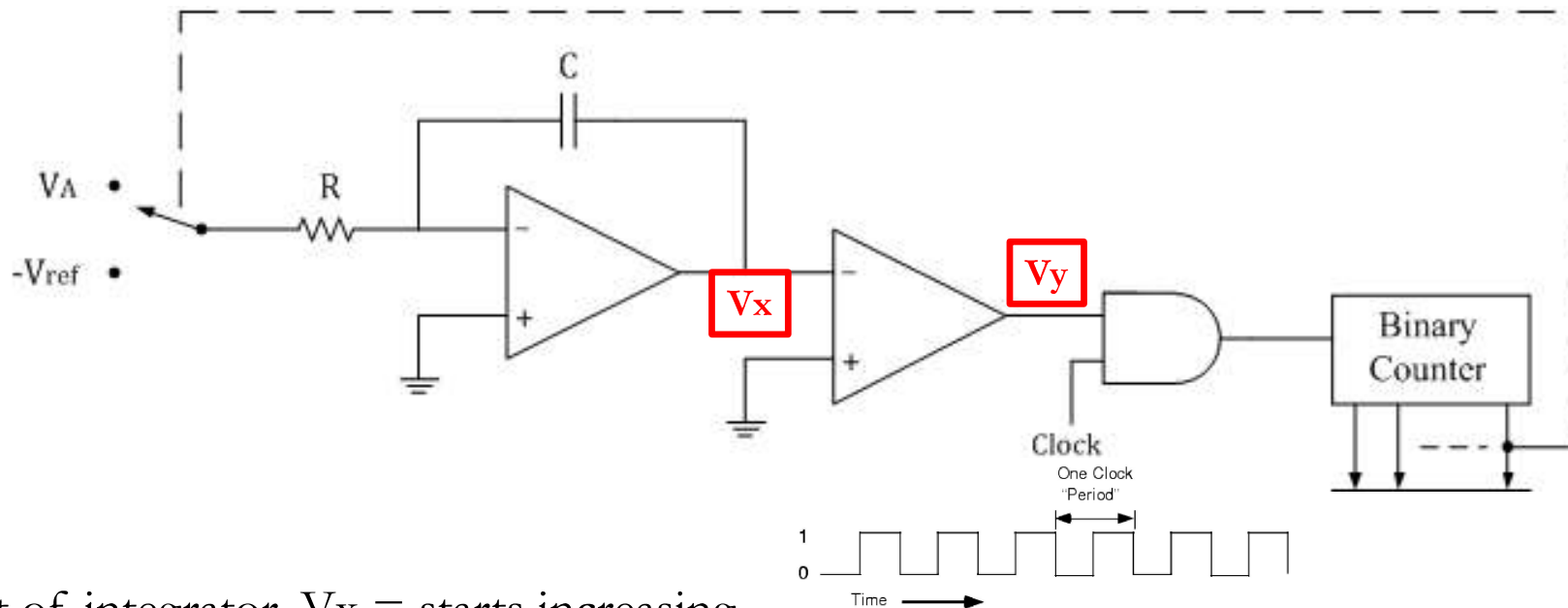


Output of integrator, $V_x = -ve$

So, the output of comparator, $V_y = "1"$

So the AND gate gives the clock pulse. And our counter starts counting from $0-2^n$. After that, the overflow bit causes the integrator to switch to $-V_{ref}$

Case 2: When connected to $-V_{ref}$



Output of integrator, V_x = starts increasing

So, the output of comparator, V_y = Becomes “0” as soon as integrator output reaches positive value.

So the AND gate output becomes “0” and the counter stops counting after T_2 time.

$$T_1 = 2^n \cdot T_c \text{ and } T_2 = N \cdot T_c$$

$$V_{in} = \frac{N}{2^n} * V_{ref}$$